PAYPAL AI HACKATHON · REFINEMENT AUDIT

The evidence intelligence layer for payment disputes

An audit of DisputeDesk AI against its positioning, a comparison with PayPal, Chargeflow and Justt, and the P0 and P1 refinements that shipped.

`Repo: rudrarajsinh920/DisputeDesk-AI` `Branch: master` `Audited at f91a960`
`P0 shipped in 7ba49a1` `P1 shipped in f08ba14` `6 Nov 2026`

Outcome

> Most of the positioning was already built. The gaps were in presentation, not logic.
>
> All three P0 refinements shipped without changes to the backend, database, sign-in, privacy, PayPal or Gemini.
>
> All four P1 refinements shipped next. They added one read-only route and one response field, with no database, sign-in, privacy or PayPal changes, and no Gemini calls.
>
> Tests after P1, run in this session: server 187 of 187 passed, client 85 of 85 passed. Typecheck and build are clean.

1. Current implementation baseline

- **Stack:** React 19 + Vite + TypeScript client; Express on Node 24 with built-in `node:sqlite`. Gemini is called only from the server.
- **Modules:** Merchant Evidence Intelligence, Buyer Dispute Clarity and Insights are built. So are sign-in with roles, server-side sessions, and an append-only audit log.
- **Demo data:** 16 synthetic disputes: 8 item not received, 4 not as described, 2 credit not processed, 1 unauthorised, 1 duplicate. They are labelled and kept apart from PayPal sandbox data.
- **PayPal:** read-only, token plus GET. Nothing is ever submitted.
- **Tests before the change:** server 183 of 183 passed and client 78 of 78 passed, both run in this session. They use synthetic data with PayPal and Gemini mocked.

2. Existing capabilities by module

A. Evidence integrity

- Four evidence types, each graded Available, Missing or Unverified.
- Key evidence defined for three of the dispute reasons.
- Four rules that catch records contradicting each other. A contested record is downgraded and never cited as fact.
- Every finding and draft line names the record it came from, such as `order.tracking_number`.
- Every gap says that missing from the records does not show it did not happen.
- No win or outcome score anywhere.

B. Fair dispute understanding

- Buyers receive only these five fields: ID, reason, status, amount and date opened.
- Every block is labelled as from the record, general guidance, or not on record.
- Stage track, next steps, and what the buyer may provide.
- States that DisputeDesk does not decide or predict the outcome.
- Another account's dispute returns 404, exactly like a missing one. Admin is read-only.

C. Dispute learning

- A pattern needs at least 3 disputes.
- Percentages are shown only from 10 disputes up.
- A monthly view needs 6 dated disputes across 3 months.
- Each recommendation lists its dispute IDs, a confidence level (moderate or limited) and a limitation.
- AI wording that uses prevent, reduce, avoid or win is rejected.

Gemini grounding

- The rules run first, and their statuses are final.
- Gemini sees only the listed record values, and every line it writes must cite them by ID.

- The server removes any line with invented numbers, dates, names or carriers; delivery, signature or pickup stories; claims that something was not shipped or not delivered; or talk about submission.
- Everything the rules found missing or in conflict is always carried into the AI draft.
- With AI off, the rule-based draft is shown and the reason is given.
- A built-in validation example runs a deliberately wrong answer through the same checks without calling Gemini.

3. Gaps against the positioning

1. **No readiness summary for each dispute.** The rules produced all the findings, but nothing summed them up. `FIXED IN P0`
2. **The buyer's claim and the records were not shown together.** The claim was a quote in a separate block. `FIXED IN P0`
3. **Positioning copy didn't say that evidence means record flags, not documents.** `FIXED IN P0`
4. **Credit not processed and duplicate disputes have no key evidence defined.** Every item counts as supporting and the draft is generic. `FIXED IN P1`
5. **Text the merchant types into the draft is never checked.** Only AI output goes through the checks for invented facts. `FIXED IN P1`
6. **Insights implies, but never states, that co-occurrence is not cause.** `FIXED IN P1`

4. Competitive comparison

Sources are official vendor pages only, as they read on 6 November 2026. "Not stated" means it was not on the pages checked. It does not mean the feature is absent.

PRODUCT	STATED ON OFFICIAL PAGES	NOT STATED ON PAGES CHECKED
PayPal disputes API developer.paypal.com	List and show disputes, provide evidence, provide supporting information, accept claim, make offer, appeal, message the other party.	Evidence guidance, flags for missing or conflicting evidence, AI.
Chargeflow chargeflow.io · /automation	Collects and enriches data automatically, and its AI “submits it on your behalf”. Claims “up to 80% higher win rates”. Fee only on recovered chargebacks. Alerts and prevention products.	Flags for missing or conflicting evidence, buyer explanations. Review is only partly stated: an option to delay submissions for internal review.
Justt justt.ai · /platform	Automatic evidence-backed arguments; a fully automated platform; recommends fight or accept partly on likelihood of success; dashboards and predictions; pay only on a win.	Flags for missing or conflicting evidence, human review of each response. Its "Consumer Clarity" works before a dispute is filed, not during one.

Not verified: PayPal's seller Resolution Center help pages. One returned a 404, one a 403 and one showed an unrelated article. The PayPal row therefore covers its public API only. Win-rate figures are the vendors' own marketing claims and were not checked.

5. Strongest defensible differentiator

An evidence check that refuses to fill gaps, traces every statement to a record, and has a person review every response.

Chargeflow and Justt both present automatic submission paid by outcome. DisputeDesk does the reverse:

- It never submits anything.
- It never scores win chances.
- It removes any AI line it cannot trace to a supplied record.
- It shows the buyer the same facts, in plain language, without the merchant's evidence.

How the current build shows it: the validation example runs a planted answer through the live checks without calling Gemini. The planted answer contains a signature story, an invented UPS tracking number, an invented date and name, and an injection attempt. All of them are removed, the honest lines pass, and the merchant sees exactly what was removed and why.

This is a positioning claim based on what the products state publicly. It does not say no competitor can do this.

6. Refinements and priority

PRIORITY	REFINEMENT	SCOPE	STATUS
P0	Evidence readiness summary on each merchant dispute: four counts from the rules, with no score.	Client only	SHIPPED
P0	Claim and records view: the buyer's claim, what the records show, and what they leave out. It assigns no fault.	Client only	SHIPPED
P0	Positioning copy (landing chip, README, Devpost tagline, demo script), stating that records are checked, not documents.	Copy	SHIPPED
P1	Refund ID guidance for credit not processed and duplicate disputes. No existing evidence type is marked key.	Evidence engine	SHIPPED
P1	Route that checks merchant-edited draft text with the existing checks for invented facts. It stores nothing and goes in the role table.	Backend	SHIPPED
P1	Readiness chip in the dispute list.	Backend (list payload)	SHIPPED
P1	Insights line: counts show what appears together, not what causes disputes.	Server text	SHIPPED

PRIORITY	REFINEMENT	SCOPE	STATUS
P2	Inspect the evidence documents themselves.	New capability	ROADMAP
P2	A refund-record evidence type.	Data model	ROADMAP
P2	Read-only PayPal sandbox evidence (GET only).	PayPal read	ROADMAP
P2	Compare the buyer's free-text claim against the records.	New rules	ROADMAP

7. What shipped in 7ba49a1

This is how the readiness summary renders for demo dispute DEMO-1002. The order is marked delivered, but there is no proof of delivery.

DEMO-1002 · Item not received — Synthetic demo record

Key evidence on record	Missing or unverified	Conflicting records	Gaps to fill in the draft
1 of 2	1	1	4

Counts from the evidence rules below. This is not a score and does not predict the outcome.

Claim and records, same dispute

- **Buyer's claim:** “The tracking page says delivered, but nothing arrived at my door.”
- **What the records show:** tracking number, order confirmation, shipping tracking and customer communication, each with its record path. The disagreement appears as "Records disagree" and is not stated as fact.
- **What the records leave out:** proof of delivery, with "Missing from the records does not show that it did not happen."
- **Close:** "It does not decide who is right; the payment provider reviews the dispute."

Files changed

FILE	CHANGE
`client/src/analysis.ts`	New `readiness()`, which counts directly from the rules' analysis.
`client/src/AnalysisPanel.tsx`	New `ReadinessSummary` and `ClaimAndRecords`, shown at the top of the merchant analysis.

FILE	CHANGE
`client/src/DisputeDetail.tsx`	Two new in-page links. The separate customer-message block is gone because its quote moved into Claim and records.
`client/src/styles.css`	Styles for the readiness cards and the three columns, using existing tokens. Two cards per row on phones.
`client/src/ui.test.ts`	Four new tests (listed below).
`client/src/Landing.tsx`, `README.md`	Positioning line and the statement that records are checked, not documents.

The Devpost draft and demo script in `reports/` were updated but are not committed. They are untracked by choice.

7b. What shipped in f08ba14 (P1)

REFINEMENT	WHAT IT DOES	GUARDS
Insights limitation	Adds: "These counts show which gaps appear in the same disputes. They do not show why any dispute was opened."	The rules' own wording runs through the same cause, trend and outcome checks as AI wording.
Refund guidance	Credit not processed and duplicate disputes get a next step pointing to proof of refund (the refund ID), which PayPal's dispute reasons reference lists for these reasons, and a specific gap in the draft asking for it.	No existing evidence type is marked key, and nothing states that a refund happened. The AI draft always keeps the gap. A refund-record evidence type stays on the P2 roadmap.
Check my edits	New merchant-only route `POST /api/analysis/check`. It runs the merchant's edited draft through the same checks as AI text and lists flagged paragraphs with reasons.	Advisory only: nothing is removed, stored or audited, and Gemini is never called. Another account's dispute answers 404. Text over 8,000 characters is refused. The route is in the role table.
List readiness	`GET /api/disputes` returns the rules' counts for each listed demo dispute. Each row shows them in words, such as "Key evidence 1 of 2 · 1 conflict".	Computed after the access filter. Dispute records are unchanged. No counts for buyers or PayPal sandbox rows, and no score or outcome words.

In a browser check as Demo Merchant A, the sentence "We shipped it with UPS, tracking 1Z999AA10123456784, on 2 October." typed into the DEMO-1001 draft was flagged for the tracking number, the date and the carrier. The rule-based lines and a sentence naming the order ID on record were not flagged. The list rows read "Key evidence 0 of 2" and "Key evidence 1 of 2 · 1 conflict". At 375px wide there was no sideways scroll.

`countMarkers` and `readiness` moved to `shared/readiness.ts`, so the server and the browser use one definition. Flag reasons reuse the AI-check wording, so some read as technical.

8. Evaluation

Tests added (passing)

- Readiness counts agree with the rules for all 16 demo disputes and the sandbox examples. No score or outcome words appear in them.
- Claim and records quotes each buyer message, never shows a contested record as fact, and uses no fault words.
- The open dispute links to both sections and quotes the buyer once.
- The buyer screen never shows either view, or the merchant-side copy of the buyer's message.
- P1: refund guidance for both reasons, with no refund stated and other reasons unchanged. The AI draft keeps the refund gap.
- P1: the edit check by role, the cross-account 404, invented values flagged, quoted buyer lines and on-record values not flagged, and oversized text refused.
- P1: list readiness equals the rules for every listed dispute, covers only disputes the account can see, and buyers get none.
- P1: the Insights limitation is present and passes the same wording checks.

Test cases by scenario

SCENARIO	EXPECTED BEHAVIOUR	COVERAGE
Missing evidence	Status is Missing. The "does not show it did not happen" line is shown, and readiness counts the gap.	Existing and new tests
Conflicting evidence	DEMO-1002: one conflict. Shipping status is not cited, and the CONFLICT marker appears in the draft.	Existing and new tests
Unsupported claims	The validation example removes every planted line and keeps the honest ones.	Existing tests
Valid evidence	All key items Available with no conflicts. Readiness shows all key evidence on record.	Partly: no demo record has every key item; a dedicated test is proposed
Too few disputes	Under 10 disputes, counts only. Under 3, no pattern. Confidence shows as limited.	Existing tests
Privacy boundaries	Buyers get five fields only. Another account's dispute returns 404. Admin cannot write.	Existing and new tests
AI turned off	The rule-based draft still appears, with the reason AI is off.	Existing tests
Synthetic-data label	The demo label is on every draft, the Insights page and the workspace banners.	Existing tests

Measurable criteria

- 100% of AI draft lines either cite a supplied record or are removed.
- Buyer responses contain zero merchant-only fields.
- Readiness counts match the rules on 16 of 16 demo disputes.
- Insights and readiness contain zero outcome words.
- Every planted line in the validation example is caught.

These measure behaviour on synthetic data. No customer impact, win rate or recovery has been measured or is claimed.

9. Risks and regression guards

RISK	GUARD
The readiness summary reads as a score	Counts only, with a stated disclaimer. A test fails on %, score, likely, chance, odds, win, lose or predict.
Claim and records reaches buyers	It is a merchant-only component, and the buyer render is tested for its absence.
The summary drifts from the rules	It is computed from the same analysis object, and tested on every demo record.
Landing copy overclaims	The existing landing test bans certified, compliant, guarantee, prevent, win rate, percentages and "trusted by". It passes.

Not checked visually: light mode, the admin view of the new panels, and the buyer screen in a browser (tests cover the buyer screen).

10. Already built vs proposed

Built and committed

- The rules-based evidence engine, contradiction rules and record citations.
- The Gemini grounding checks, the AI-off fallback and the validation example.
- The buyer view, with every block labelled by where it comes from.
- Insights sample-size thresholds and confidence levels.
- Evidence readiness and Claim and records (7ba49a1).
- Positioning copy in the landing page and README (7ba49a1).
- P1: refund ID guidance, Check my edits, list readiness counts and the Insights limitation (f08ba14).

Proposed, not built

- P2: document inspection, a refund-record type, sandbox evidence reads, and buyer-claim comparison.

All dispute data shown is synthetic demo data, not real PayPal disputes. No live Gemini or PayPal calls were made for this audit or its refinements. Competitor statements come from the official pages linked above, as they read on 6 November 2026.